R-16

MIL-O-43085C

13 December 1976

SUPERSEDING
MIL-O-43085B
11 March 1971

MILITARY SPECIFICATION

OVEN, BAKING AND ROASTING, DECK, GASOLINE, 3-COMPARTMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of gasoline-heated, 3-compartment, baking and roasting oven.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

HH-I-558	-	Insulation, Blocks, Boards, Blankets, Felts, Sleeving, (Pipe and Tube Covering), and Pipe Fitting Covering, Thermal (Mineral Fiber, Industrial Type).
QQ-A-225/8	-	Aluminum Alloy Bar, Rod, Wire and Special Shapes; Rolled, Drawn or Cold Finished, 6061.
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet.
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet.
QQ-I-666	-	Iron, Malleable, Ferritic, for Castings.
QQ-S-766	-	Steel Plates, Sheets and Strip - Corrosion-Resisting.
QQ-W-423	-	Wire, Steel, Corrosion-Resisting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

95

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:30Z
Check the source to verify that this is the current version before use.

UU-P-268 - Paper, Kraft, Wrapping.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-C-11133 - Crates, Shipping, Wood, Open, Wirebound.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS 35791 - Handle, Bail, Chest, Surface Mounting, Countersunk Holes.

DRAWINGS

ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-11-738 - Oven, Baking and Roasting, Deck, Gasoline, 3-Compartment; Assembly.
5-11-739 - Sections.

(Figure 1, this figure is a miniature copy of the issue of Drawing 5-11-738 in effect on the date of specification approval, and is for information only. The issue of the drawing referenced in the contract shall govern.)

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AMD MATERIALS (ASTM)

A 575 - Merchant Quality Hot-Rolled Carbon Steel Bars

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

95

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the currentversion before use.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies of Handbook H28 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.2, 6.2, and 6.3).

3.2 Materials and components. Materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for his baking and roasting ovens, provided the completed item complies with all provisions of this specification.

3.2.1 Steel, corrosion-resisting.

3.2.1.1 Sheet, strip, and shapes. The corrosion-resisting steel sheet, strip and shapes shall conform to class 302 or 304 of QQ-S-766.

3.2.1.2 Rod. Corrosion-resisting steel rods used for stays shall conform to form I, condition A, composition 410, 420 or 430 of QQ-W-423.

3.2.1.3 Wire, spring. Corrosion-resisting steel wire used for springs shall conform to form I, condition B, composition 302 or 304 of QQ-W-423.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:30Z
Check the source to verify that this is the current version before use.

3.2.2 Carbon steel.

3.2.2.1 Pipe. Steel pipe shall be standard weight, galvanized.

3.2.2.2 Rod. Carbon steel rod shall conform to ASTM A 575.

3.2.3 Aluminum.

3.2.3.1 Alloy 5052. The 5052 aluminum alloy shall conform to QQ-A-250/8. Temper shall be as specified on the drawings.

3.2.3.2 Alloy 6061. The 6061 aluminum alloy used for the structural frame shall conform to temper T6 of QQ-A-225/8.

3.2.3.3 Alloy 3003. Aluminum alloy 3003 shall conform to temper H14 of QQ-A-250/2.

3.2.4 Iron, malleable. The malleable iron shall be grade IG of QQ-I-666, except that identification marking requirements are not required.

3.2.5 Insulation. Insulation shall be blanket-type, conforming to form B, type 1, class 8, of HH-I-558.

3.2.6 Hardware. Hardware shall be the manufacturer's standard commercial items. Unless otherwise specified on the drawings, all hardware shall be fabricated of corrosion-resisting steel. Threads shall conform to National Bureau of Standards Handbook H28.

3.2.7 Thermometer. The thermometer shall be industrial, dial type with a temperature range of 50 degrees Fahrenheit (F.) to 500 F. The casing shall be corrosion-resisting steel, 2 inches in diameter, with a glass window designed for easy reading. All other parts shall be metal and the scale plate shall be permanently marked with black figures. An 18 inch stem with a nominal 1/4-inch American Standard Taper Pipe Thread (NPT) connection nut shall be provided.

3.2.8 Handle. Handles shall conform to MS 35791-1.

3.3 Design and construction. The construction of all components and assemblies of the item covered by this specification shall conform to the requirements specified herein and the design, details, dimensions, material classification and sizes shown on Drawing 5-11-738, and supporting drawings referenced thereon.

3.3.1 Insulation assembly. Insulation shall cover the required area and shall contact all edges. The insulation for the sides and top shall be one

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:30Z
Check the source to verify that this is the current version before use.

continuous piece. The insulation covering the rear panel shall be of one piece and shall be wire stitched at the top and sides connecting it to the insulation covering the sides and top.

3.4 Marking.

3.4.1 Instructions. The instructions shown on Drawing 5-11-753 shall be engraved or stamped on the instruction plate and filled with black heat-resisting enamel paint.

3.4.2 Identification. Identification marking shall be stamped or engraved on the identification plate in characters of the size shown on Drawing 5-11-753 and filled with black heat-resisting enamel paint.

* 3.5 Finish. All finishes shall be free of rust, stain, and shall have smooth surfaces and edges.

3.5.1 Galvanizing. Galvanized finish for components indicated on drawings shall be accomplished by either the hot dip or electrodeposited method in accordance with standard commercial practice.

3.5.2 Exterior finish. All exposed corrosion-resisting steel surfaces of the cabinet shall have a number 2D finish. Aluminum alloy used in the fabrication of the base and side frame bars, rivets, and aluminum sheet used in the fabrication of sides and back shall have a dull mill finish.

3.5.3 Interior finish. All visible interior corrosion-resisting steel surfaces shall have a number 2D finish, and all visible interior aluminum alloy surfaces shall have a dull mill finish.

* 3.6 Workmanship. All components and assemblies of the range shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges and sharp corners. Functioning component shall function as intended. Adjustable component shall function without binding. No component of subassembly shall be misaligned. Door(s), door stay rod(s) and rod-hinges shall not be bent. Sheet metal or tubing shall be free from dents and scars. Components shall not be fractured, buckled, warped, loose, dented, punctured or malformed.

* 3.6.1 Metal fabrication. Metal used in the fabrication of the range shall be free from kinks. Forming and shearing shall not cause damage to the metal and shall be free of trim marks.

* 3.6.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:30Z
Check the source to verify that this is the current version before use.

of cracks, burn holes, undercuts or incompleted fushion. All scale and flux shall be removed from the finish weld area.

* 3.6.3 Fastening devices. Threaded fastener and rivet holes shall have burrs removed. Threaded fasteners shall not be missing or broken, cracked or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection. When a first article is required, the oven shall be examined for defects listed in table I and for compliance with dimensions specified. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the first article.

4.3 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection. Examination shall be made of insulation installation as specified in 3.3.1 to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the item affected and to the operation.

4.3.3 End item inspection. The inspection lot shall consist of all ovens offered for inspection at one time. The sample unit shall be one completely fabricated oven.

* 4.3.3.1 Visual examination. The oven shall be examined for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Stainless steel not No. 2D finish	X	
	Not galvanized (where required)	X	
Construction and workmanship (general)	Part missing or not as specified	X	
	Fractured, buckled, warped, loose, dented, punctured, or malformed	X	
	Not in proper alignment	X	
	Any sharp edges or sharp corners	X	
	Any functioning component that is inoperative or will not operate as intended.	X	
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended	X	
	Any component not mounted in the manner specified on applicable drawings	X	
Pipe and tubing	Not type or material specified	X	
	Improperly connected or installed; dented or scarred	X	
Welding	Not continuous, sound, smooth, free from rust, scale, paint, grease and other foreign matter	X	
	Any cracks, burn holes, incomplete fusion, or undercuts	X	
	Scale or flux deposits not removed		X
Fastening devices	Missing	X	
	Bolts, screws or other threaded fasteners, stripped, fractured or loose		X
	Rivet(s) loose, broken, not peened, or insufficiently peened	X	
	Rivet heads not in full contact with riveted members	X	
Assembly	Component not properly assembled or secured, e.g., lock washer missing from metal fastener		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:30Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Marking identification	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner, not correct color	X	
Instructions	Missing, incomplete, not legible	X	

4.3.3.2 Dimensional examination. The oven shall be examined for compliance with dimensions specified. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.4 Packaging inspection. An examination will be made to determine that preservation-packaging, packing and marking as required by section 5 of this specification are complied with. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-4 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

TABLE II. Examination of packaging

Examine	Defect
Marking, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete packaging of draft diverter or thermometer or securing of lifting handles. Blocking, bracing or cushioning inadequate or missing.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or commercial as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the current version before use.

* 5.1.1 Level A. The draft diverter shall be removed from the top of the oven and completely wrapped in minimum 40-pound kraft paper conforming to type I, grade B of UU-P-268. The kraft paper wrap shall be secured with gummed paper tape minimum 2 inches in width conforming to type III, grade B of PPP-T-45. The wrapped diverter shall then be placed in the top compartment of the oven and blocked, braced, and cushioned to prevent movement while in transit or storage. The three oven doors shall be closed and secured in place. The thermometer face shall be protected with single face corrugated fiberboard and the draft diverter opening covered with waterproofed kraft paper. All crevices around the doors, the covered thermometer including the guard, and the covered draft diverter opening shall be sealed with tape conforming to PPP-T-60. The two lifting handles shall be wired, taped or tied securely in place to prevent movement while in transit or storage.

* 5.1.2 Commercial. The ovens shall be preserved-packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The contractor may use his standard commercial practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B packing. Each oven, preserved-packaged as specified in 5.1, shall be packed in a crate constructed in accordance with the requirements ot MIL-C-11133 and as shown on Drawing 5-11-755.

5.2.2 Commercial packing. Each oven, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. One slat on each of the four vertical faces of each shipping container shall be marked in characters and symbol not less than 2 inches in height as follows:



6. NOTES

6.1 Intended use. The item covered in this specification is used in conjunction with the burner unit covered by MIL-B-40098 for roasting and

Source: https://assist.dla.mil -- Downloaded: 202640-21T20:30Z
Check the source to verify that this is the currentversion before use.

baking food for troop field operation. Both items are components of MIL-B-43322: Bakery Plant, Field, Portable, Consisting of 170 Components.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1, 4.2, and 6.3).
(c) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one oven. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - MC

Review activity:

Army - MD
Navy - YD
DGSC - GS

Preparing activity:

Army - GL

Project No. 7310-0334

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the current version before use.

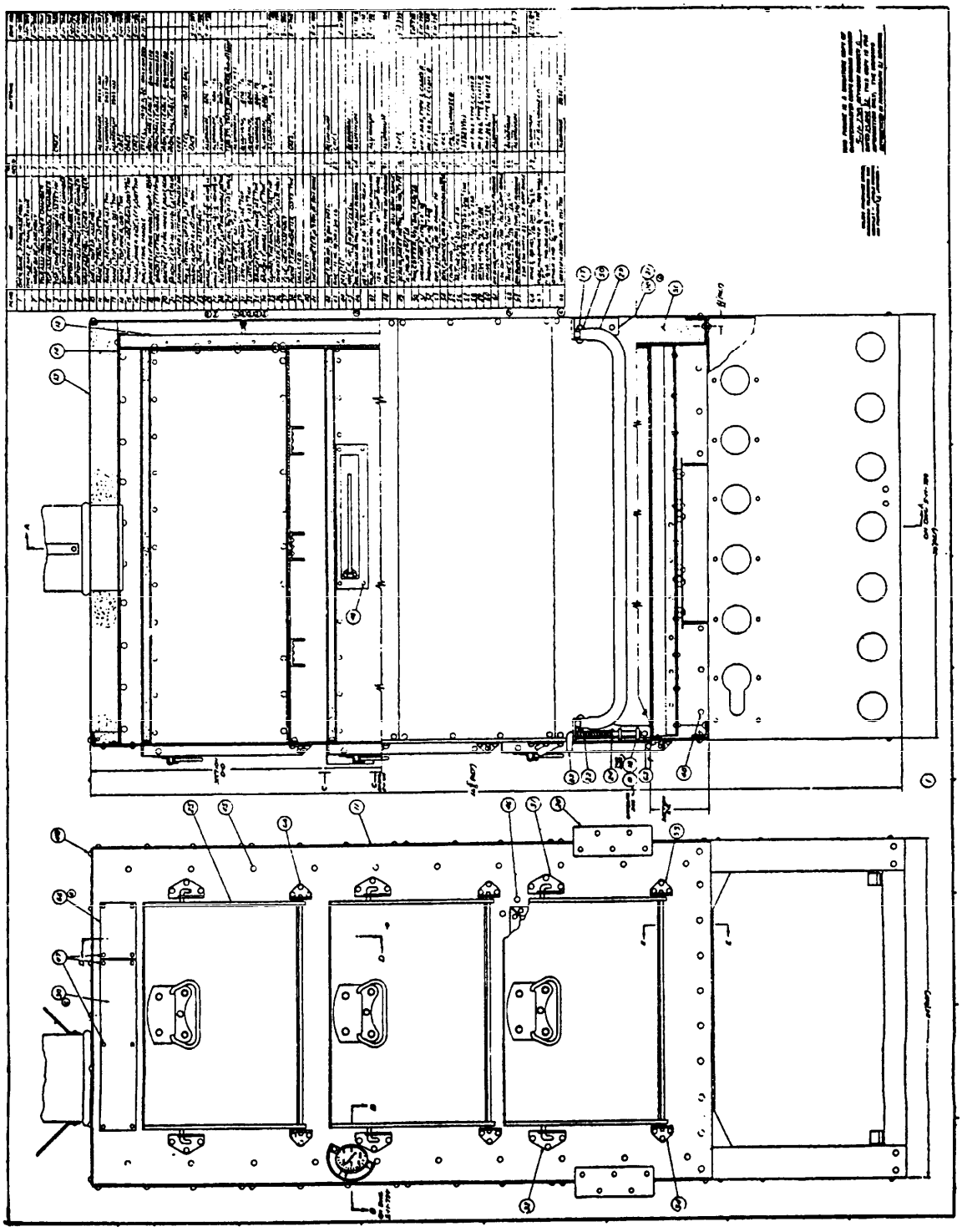


FIGURE 1. Oven, baking and roasting, deck, gasoline, 8 compartments; assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DOD 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Natick Research and Development Command
Attn: DRXNM-TD
Natick, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:30Z
Check the source to verify that this is the current version before use.